

FRONTONE, Italy

WMO: 161790

Lat: 43.517N

Lon: 12.728E

Elev: 1883

StdP: 13.72

Time Zone: 1.00 (EUC)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.9	28.4	10.4	10.0	36.7	14.4	12.2	36.8	38.5	48.4	32.6	48.6	6.8	250	0.439

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	93.1	67.3	89.9	66.8	87.1	66.2	71.4	84.2	69.8	82.8	68.4	81.5	8.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.8	109.7	77.2	65.6	101.5	75.6	63.9	95.5	74.3	36.4	83.7	35.0	82.4	33.8	81.3	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.1	22.2	17.9	DB	22.9	97.3	3.4	3.5	20.4	99.8	18.4	101.8	16.5	103.8	14.0	106.4
			WB	20.6	74.9	2.9	1.7	18.5	76.1	16.8	77.1	15.2	78.1	13.1	79.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.2	40.5	41.0	47.1	55.1	62.2	71.4	76.7	75.8	65.9	57.7	49.2	42.7
	DBStd	14.41	6.61	7.19	7.58	5.93	7.07	7.58	6.09	6.39	6.75	6.51	6.50	6.35
	HDD50	1067	300	257	147	18	2	0	0	0	0	14	91	238
	HDD65	3933	760	671	556	299	142	27	3	3	67	238	475	692
	CDD50	3694	5	6	56	173	380	642	826	799	478	252	66	11
	CDD65	1084	0	0	0	3	56	219	364	337	94	11	0	0
	CDH74	8505	0	0	0	8	280	1729	3260	2796	420	12	0	0
	CDH80	3127	0	0	0	0	50	585	1294	1102	96	0	0	0
Wind	WSAvg	6.3	7.0	7.2	7.2	6.1	5.8	5.6	6.0	5.5	5.6	5.2	6.7	7.4
Precipitation	PrecAvg	47.2	3.2	3.7	3.7	4.2	4.0	3.4	2.6	3.5	4.1	4.6	6.1	4.3
	PrecMax	73.8	6.9	8.1	7.0	9.8	10.7	6.9	8.4	12.7	8.6	10.6	13.5	7.7
	PrecMin	25.7	0.3	0.8	0.6	0.9	0.1	0.7	0.5	0.1	0.8	0.0	2.2	0.8
	PrecStd	10.9	2.0	2.4	1.9	2.2	2.4	1.8	1.9	2.9	2.7	3.0	2.5	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.6	59.2	67.4	75.1	84.5	93.4	96.8	97.9	87.6	75.5	64.6	58.5
		MCWB	46.0	46.3	51.3	55.7	63.8	68.2	66.9	67.0	64.3	60.2	57.7	52.3
	2%	DB	53.9	55.4	63.5	71.1	80.5	89.8	93.3	93.3	82.6	71.8	61.9	55.7
		MCWB	47.0	45.8	50.4	55.7	62.9	67.8	67.3	66.9	64.7	60.8	55.7	50.3
	5%	DB	51.9	52.5	60.1	67.9	76.9	86.3	90.1	89.5	78.7	68.4	59.5	53.5
		MCWB	46.3	45.4	49.4	53.8	61.6	66.4	67.1	66.2	63.2	59.5	54.0	48.7
	10%	DB	49.8	50.2	57.1	64.5	73.6	82.9	87.4	86.1	75.3	66.1	57.4	51.4
		MCWB	45.0	44.5	47.9	52.6	59.4	65.0	66.4	65.5	62.3	58.9	52.4	47.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.1	51.0	54.8	61.5	68.5	73.4	74.4	72.6	69.6	65.0	61.2	53.8
		MCDB	53.6	54.8	61.7	68.6	79.1	85.3	86.0	85.1	79.2	69.7	62.7	56.9
	2%	WB	48.4	48.8	52.6	58.0	65.2	70.5	71.4	70.6	67.2	63.2	57.5	51.2
		MCDB	52.6	52.3	59.3	65.8	76.3	83.3	84.8	83.1	76.1	68.1	60.0	54.5
	5%	WB	47.0	47.1	50.9	55.9	62.8	68.6	69.9	69.2	65.7	61.4	55.1	49.3
		MCDB	50.9	50.4	57.2	63.6	73.4	82.0	83.8	81.9	74.0	65.9	57.9	52.5
	10%	WB	45.4	45.4	49.3	54.3	60.5	66.6	68.4	67.8	64.2	59.8	53.1	47.4
		MCDB	49.0	49.0	55.4	61.4	71.2	80.3	82.0	80.9	72.4	64.1	56.4	50.6
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.2	10.8	13.4	14.9	16.0	16.9	16.4	13.2	9.9	7.8	7.2
		MCDBR	9.1	10.6	14.2	16.6	17.6	18.3	19.1	18.9	16.7	12.4	9.5	8.3
		MCWBR	6.6	7.3	8.3	9.8	10.9	10.3	9.5	8.9	8.2	7.3	6.9	6.7
	5% WB	MCDBR	8.4	8.9	12.4	14.7	16.7	17.1	17.0	16.6	14.4	10.8	8.4	7.6
MCWBR		6.8	7.4	8.2	9.6	11.1	10.4	9.7	9.3	8.6	7.3	7.2	6.9	
Clear-Sky Solar Irradiance	taub	0.320	0.339	0.368	0.401	0.403	0.403	0.395	0.392	0.381	0.366	0.342	0.318	
	taud	2.460	2.386	2.318	2.236	2.272	2.308	2.330	2.355	2.375	2.411	2.421	2.483	
	Ebn at Noon	257	270	275	274	276	275	276	273	268	256	244	246	
	Edh at Noon	23	30	36	42	42	40	39	37	34	29	24	21	
All-Sky Solar Radiation	RadAvg	453	709	1085	1454	1804	2086	2163	1868	1328	858	512	407	
	RadStd	62	94	154	143	167	110	108	129	89	89	56	55	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)	+0.68	N/A	N/A	N/A	N/A	N/A	N/A	-139	+149	N/A

Nomenclature: See separate page